APERIO | AUTONOMOUS EARTH INTELLIGENCE PLATFORM

AUTHORITY DISASTER INTELLIGENCE ASSESSMENT • DEFORESTATION

Location: Forest Grid (10.89N, 76.52E), Kerala, India • Generated: 09 Oct 2026 04:15 UTC • Status: AWAITING HUMAN REVIEW

Target Event	Deforestation • Forest Grid (10.89N, 76.52E)	**Risk Probability**	**1% (HIGH)**
Affected Area	Survey extent	**Model Confidence**	80% Indicative
Exposed Population	Assessed in loop	**Antecedent Rainfall**	120 mm (72h: 572mm)
Terrain Slope	15° (Steep escarpment)	**SAM Scar Boundary**	Validated (7.03 km²)
Satellite Observation	Sentinel-1 SAR + Sentinel-2 MSI	**Human Verification**	AWAITING HUMAN REVIEW

Multi-Agent Intelligence & Evidence Progression

[SYNTHETIC DEMO DATA NOTICE] This report record is derived from synthetic demonstration data (*aperio_synthetic_deforestation_dataset.csv*) and does not represent verified satellite ground truth.
1. Satellite Optical Ingestion: Simulated Sentinel-2 multi-spectral canopy reflectance with NDVI change calculation.
2. SAM Segmentation: Simulated Zero-Shot Segment Anything canopy loss mask.
3. XGBoost & Bayesian Sequential Update: Optical disturbance and SAR backscatter features updated beliefs to calibrated probability rating of **1%**.

Conservation Advisory & Ground Inspection Planning

Designation	Unit / Command	Distance / Status	Exposure
State Forest Division Monitoring Post (Kerala)	Forestry Range Unit	1.5 km • Patrol Track Open	**MONITORING**
Forest Grid (10.89N, 76.52E) Buffer Perimeter Station	Conservation Squad	3.2 km • Operational	**SURVEILLANCE**

Satellite Change Intelligence & Extent Quantification

Analyzed Extent	12056.04 km² (Full Granule)	**Methodology**	Visual Green Canopy Disturbance Delta
Baseline Affected Extent	0.45 km² (2024-02-07)	**Sensor & Product**	Sentinel-2 MSI Level-2A (Tile 43PFM)
Latest Affected Extent	486.38 km² (2024-05-07)	**Cloud Cover**	Pre: 0.36% • Post: 7.42%
Newly Affected Area	**485.93 km²**	**Net Change**	**485.93 km²** (+107984.4%)
Detection Zones	6 Delineated Zones	**Integrity Status**	**CANDIDATE_CHANGE**

Accountable Operator Review & Audit Trail

Reviewing Desk: APERIO Response Operations Desk
Operator Notes: Automated verification report
Operational Boundary: Decision Support Output only. All evacuations require local ground confirmation.